

COPELAND ROYALL

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SKILLS

Languages Python, TypeScript, Go

Software AWS, GCP, Terraform, Terragrunt, Pulumi, Ansible, GitHub Actions, Docker, Kubernetes

EXPERIENCE

Freelance Software Engineer

January 2025 – present

Opinion Health

Architected and built pharmaceutical research AWS environment using Terraform and Terragrunt

- Multi-region account-isolated RDS Aurora for data residency of healthcare data
- Dedicated log archive account for org CloudTrail and region-matched pgAudit database logs exported via Lambda with industry-compliant 6-year retention
- Delegated-account administration of org-wide threat detection (GuardDuty) and guardrails (SCPs for region locking, encryption and KMS)
- GitOps using GitHub Actions CI/CD pipeline with Terraform module unit testing and security/secret checks, utilized in PR review workflow and drift detection
- Python NLP pipeline for extreme multi-label classification (1.4k labels) on 130k patient records spanning 10 years using traditional techniques, Bedrock embeddings and batch LLM inference; users anonymized until database ingest to reduce exposure risk
- SSO+MFA IAM database access from Ansible-controlled VMs
- Lambda for preemptive cold start of Aurora database to ensure initial queries don't timeout

Keygraph

- Developed multi-account, -region and -environment AWS platform module in TypeScript/Node.js Pulumi IaC
- Deployed 6 containerized production Go apps using App Runner, RDS Aurora and SOPS secrets

Historical Artist Website

- Built website for acclaimed 20th-century San Francisco artist with Hugo, Cloudflare Worker and R2 deployment; JavaScript OAuth proxy for client editing

Visa Software Engineer

September 2023 – May 2024

Accelerated EU payment product rollouts on GCP with TypeScript IaC and led deployment of cost monitoring in org.

- Implemented TypeScript Pulumi IaC package deploying 256 resources across 4 GCP projects, meeting PCI compliance requirements
- Wrote reusable Pulumi packages for GKE, IAM, and KMS resource management with comprehensive TypeScript unit test coverage
- Led team initiative to deploy Cloudability FinOps in 45 test and production projects; calculated 10% cost savings
- Installed and updated Kubernetes cluster addons using Argo CD, Kustomize and Helm
- Migrated Concourse CI pipelines to new load-test platform

Visa DevOps Engineering Intern

Summer 2022

Developed foundational Terraform modules for AWS RDS Aurora and ElastiCache Redis including best-practice defaults to enforce consistency across teams.

Tangible AI Natural Language Processing Intern

Summer 2021

Trained a TF-IDF/ridge-regression model (R^2 0.61) for a positivity-ranked personalized news reader feature for chatbot.

EDUCATION

University of Edinburgh BSc Computer Science, First-class honors

2019 – 2023

Dissertation: Microservices and message queue-based multi-receiver ADS-B aircraft tracking data aggregation system written in Go, utilizing Apache Kafka, MongoDB and GraphQL, achieving 10ms end-to-end latency

PROJECTS

- Home Kubernetes cluster deployed with Flux CD, Prometheus/Grafana, SOPS secrets, cert-manager, Ubuntu
- Built first 3D aircraft flight heatmap and radar viewer to see paths curving with the globe (instead of distorted 2D)
- Empowered Siri with local aircraft tracking, achieving over 18,500 downloads from GHCR